POC of a Quantum Self Supervised Semantic Segmentation Framework

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***Abstract*—While self-supervised learning (SSL) has significantly advanced unsupervised semantic segmentation by extracting rich visual features from unlabeled data, existing classical frameworks often struggle to capture complex, high-dimensional semantic correlations. At the same time, quantum image segmentation methods leverage quantum properties for efficient clustering but typically fail to inherit or maintain semantic meaning. To bridge this gap, this work introduces a hybrid Quantum Self-Supervised Semantic Segmentation (Q4S) framework. Our approach integrates a quantum segmentation head into a classical SSL architecture, exploiting high-dimensional Hilbert spaces with the aim to enrich semantic feature correlations. We validate our proof of concept on a CocoStuff subset of a 100 images, achieving a mean Intersection over Union (mIoU) of 0.58 and a Mean Pixel Accuracy (MPA) of 0.80. These results demonstrate the feasibility of embedding quantum components into self-supervised pipelines, offering a promising baseline for future research in quantum-enhanced scene understanding.**

***Index Terms*—Self Supervised Learning, Semantic Segmentation, Variational Quantum Algorithms**

I. Introduction

The vast availability of unlabelled visual data across fields such as remote sensing and medical imaging renders traditional fully supervised semantic segmentation impractical, as this approach often relies on imprecise ground truth masks [1]. To mitigate the reliance on manual annotation, self-supervised learning techniques have been developed, most notably contrastive learning. By employing advanced frameworks such as STEGO [2] and recent universal semantic segmentation methods [3], researchers can now generate superior visual representations, thereby pushing the boundaries of unsupervised scene understanding.

Recent unsupervised segmentation approaches leverage quantum mechanisms, including quantum annealing via the QUBO formalism [4], [5], [11] and quantum circuits [6], to achieve efficient pixel clustering. However, these methods rely strictly on low-level intrinsic features like color and variance, leaving them unable to capture the underlying semantic meaning of the image.

This research introduces the Quantum Self-Supervised Semantic Segmentation (Q4S) framework, a hybrid architecture designed to bridge the gap between quantum unsupervised segmentation and classical self-supervised methods. The framework consists of three sequential stages: training a classical model to align augmented images, applying a quantum segmentation head via a variational circuit to enhance semantic correlations within a high-dimensional Hilbert space, and finally feeding the extracted observables into a k-means clustering algorithm [7]. Although currently constrained by Noisy Intermediate-Scale Quantum (NISQ) technology, Q4S serves as a proof of concept demonstrating the algorithmic feasibility of quantum components in self-supervised tasks, paving the way for future Fault-Tolerant Quantum Computing (FTQC) implementations.

Contributions

Our primary contribution is the proposal of Q4S, a hybrid self-supervised architecture designed for one image at a time in order to integrate a quantum component into semantic segmentation. We developed a specific encoding methodology to adapt high-dimensional embeddings from a classical encoder to a Variational Quantum Circuit (VQC) limited to 9 qubits, complying with current NISQ constraints. We validate this proof of concept on a dataset of a hundred images, achieving promising results (Table II). These results empirically demonstrate that a variational quantum head can effectively process and cluster complex semantic representations, establishing a baseline for quantum-enhanced self-supervised learning.

II. Methods

In this section, we present the overall methodology underlying our framework. It consists of three sequential modules designed to progressively capture, refine, and exploit semantic information.

A. Self-Supervised Pretraining Stage

Self-supervised learning is a subfield of unsupervised learning that enables the extraction of discriminative representations without requiring labeled data [8].

The objective of this step is to align semantically similar images despite geometric variations. This enables the model to associate semantic meaning with each pixel, since the only shared elements between the two images are the objects they contain. Prior work has demonstrated that a trained model is already capable of assigning semantic meaning at the pixel level [2]. Consequently, our goal is to recover an encoder that

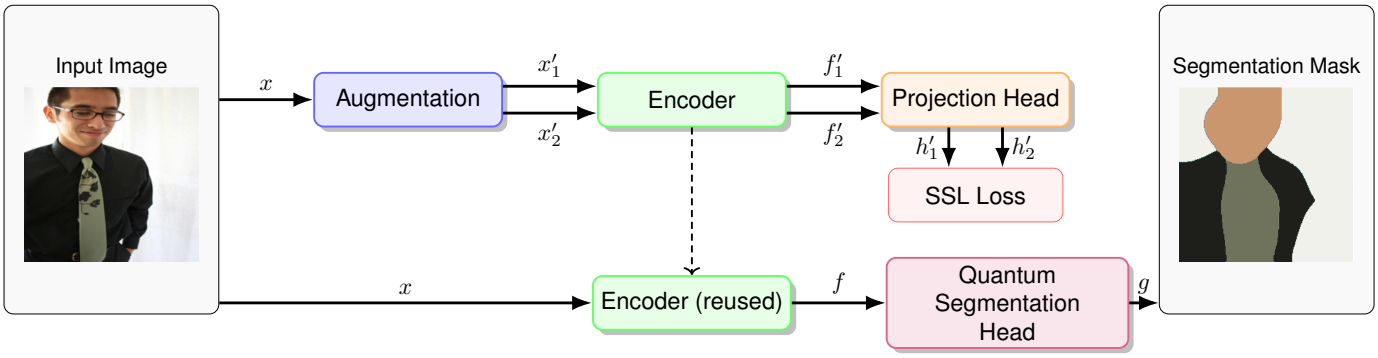


Fig. 1. General architecture of the Q4S framework. The process begins with an iterative self-supervised learning phase using a Self-Supervised Loss (SSL) function, after which the frozen encoder is leveraged to train the quantum segmentation head for clustering. Notation: x is the input image; x'_1, x'_2 are geometric augmentations; f'_1, f'_2 are backbone feature vectors; h'_1, h'_2 are projection head outputs; and g is the final quantum segmentation output.

can effectively capture pixel-level semantic information for a given image [9].

To this end, we adopt the architecture illustrated in Figure 1. Starting from an image x, we generate two augmented views, x'_1 and x'_2, by applying random crops and rotations. Both views are then processed by our encoder to produce encoded representations f'_1 and f'_2. These representations are subsequently passed through a projection head, yielding two vectors h'_1 and h'_2. Finally, the similarity between these vectors is optimized using a self-supervised loss function. In our study, we employ the *SimCLR loss* [10], a state-of-the-art objective function for contrastive learning.

As these components follow standard designs from existing literature, we omit their implementation details here. A complete specification of the encoder and projection head architectures, training hyperparameters, and optimization strategies is provided in Section III.

B. Objective - Semantic Segmentation Stage

During this stage, our objective is to strengthen the relationships among pixels that share the same semantic meaning, while discriminating those between semantically distinct pixels. To this end, we start from an input image $x \in \mathbb{R}^{C\times H'\times W'}$, where C denotes the number of channels and (H', W') the spatial dimensions. The image is processed by the pretrained encoder, yielding a feature map $f \in \mathbb{R}^{K\times H\times W}$, where K is the embedding dimension and (H, W) are the spatial dimensions of the encoded representation, with $H < H'$ and $W < W'$. The resulting feature map f is then passed through our quantum segmentation head to obtain $g \in \mathbb{R}^{N\times H\times W}$, where N denotes the number of qubits. This process is illustrated in Figure 1.

We now introduce our quantum segmentation head and the procedure for constructing the feature map g. For each spatial location $(h, w) \in \{1, \ldots, H\} \times \{1, \ldots, W\}$, we extract the corresponding feature vector of dimension K from f and feed it into the quantum segmentation head, thereby constructing one spatial region at a time, as depicted in Figure 2.

The quantum segmentation head first encodes the classical information into a quantum state using *amplitude embedding*.

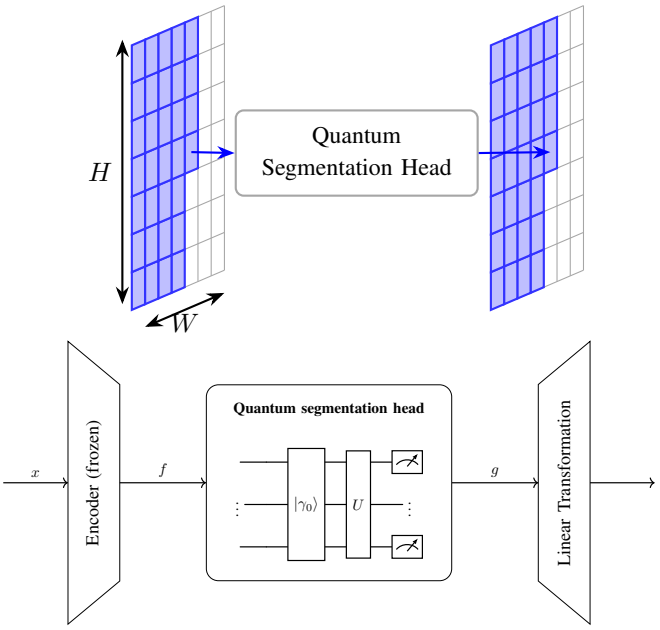


Fig. 2. Parallelization of the processing of each spatial zone of the feature vector of our backbone to generate the quantum semantic projection. Each vector is encoded using 9 qubits and processed through an Ansatz constructed from strongly entangling layers.

This technique embeds 2^N features into the amplitude vector of N qubits, which imposes the constraint $K = 2^N$. The quantum-encoded information is then processed through a sequence of strongly entangling layers. The depth was fixed to $L = 1$ layer based on empirical gradient stability rather than simulation cost alone. Preliminary trials with $L > 1$ exhibited vanishing-gradient behavior, which is consistent with barren-plateau effects reported for deep variational circuits. Importantly, quantum advantage has been supported by recent studies [12], which demonstrate that quantum properties, namely superposition and entanglement, enable the modeling of richer and more complex relationships across spatial regions. Furthermore, the numerous degrees of freedom offered by this Ansatz allow for optimally capturing the information

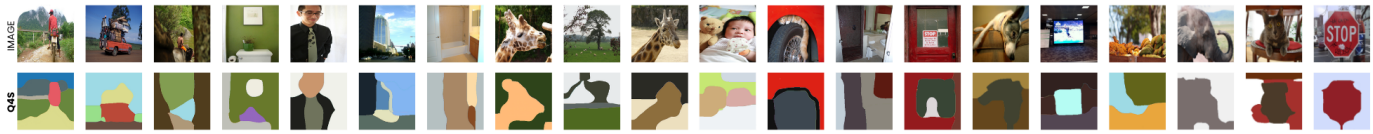


Fig. 3. Qualitative results of our Q4S framework on various images from the dataset. From top to bottom: input image and predicted segmentation mask by Q4S. The results demonstrate the framework's ability to capture semantic structures effectively.

of f. This enables a parameterization tailored to each studied image. Finally, we augment the dimension of the output observables using a classical linear transformation to align the quantum output with the required classification space.

C. Contrastive Loss for semantic segmentation

To increase inter-pixel relationships, we use STEGO's contrastive loss, modified to incorporate quantum observables. To understand our modifications, let us start from the initial loss function, defined as:

$$\mathcal{L} = \lambda\,\mathcal{L}_{stego}(x, x, b) + \lambda_{knn}\,\mathcal{L}_{stego}(x, x_{\text{knn}}, b_{\text{knn}}) + \lambda_{\text{rand}}\,\mathcal{L}(x, x_{\text{rand}}, b_{\text{rand}}) \tag{1}$$

Here, λ denotes a set of hyperparameters, and b represents various regularization parameters. The total loss consists of three terms: the STEGO contrastive loss applied first between the image x and itself, then between x and one of its nearest neighbors x_{knn} based on inter-pixel similarity (defined in Equation 6), and finally between x and a randomly sampled image x_{rand}. In our case, our framework is designed to process a single image at a time to avoid gradient loss, requiring per-image retraining ($\sim$1 min on an A100 GPU). Therefore, we retain only the first term in order to avoid extra quantum forward passes that would compound gradient instability and cost:

$$\mathcal{L} = \lambda\,\mathcal{L}_{stego}(x, x, b) \tag{2}$$

The STEGO contrastive loss is given by 3:

$$\mathcal{L}_{stego}(f, f', b) = -\sum(F(f, f') - b)S \tag{3}$$

where f and f' are the feature vectors corresponding to images x and another image respectively, S is a segmentation correlation tensor between g and g' of our quantum segmentation head. To achieve this, we define the vector of expectation values for a state ψ 4:

$$V_\psi = [\langle\hat{O}_1\rangle_\psi, \ldots, \langle\hat{O}_N\rangle_\psi]^T \tag{4}$$

To increase dimensionality ($N \to M$), we apply a learnable linear transformation matrix $W \in \mathbb{R}^{M\times N}$. Finally, the similarity metric S is computed in the projected space 5:

$$S(\psi_g, \psi_{g'}) = \frac{(WV_{\psi_g}) \cdot (WV_{\psi_{g'}})}{\|WV_{\psi_g}\|\|WV_{\psi_{g'}}\|} \tag{5}$$

We specifically chose the Pauli-Z observable for each qubit, as its expectation values are bounded within the interval $[-1, 1]$. This normalization property facilitates direct comparison of the quantum outputs, making the resulting vectors well-suited for subsequent classical analyses, such as the computation of cosine similarity between different experimental runs or data points. Finally F is the cosine similarity function defined as:

$$F = \sum_c \frac{f_{chw}\, f'_{chw}}{\|f_{hw}\|\,\|f'_{hw}\|} \tag{6}$$

A key characteristic of this methodology is that the backbone remains frozen during this stage; only the segmentation head is trained. This makes the process particularly efficient, both computationally and in terms of training time.

D. Clustering Stage

The final step of the framework generates semantic clusters by first resizing low-resolution feature vectors to the original image dimensions using bilinear interpolation, ensuring pixel-perfect segmentation that preserves both spatial and semantic information. Each pixel vector then undergoes strict normalization to allow the use of cosine similarity during the subsequent clustering process. Finally, a standard k-means algorithm partitions the pixel vectors into disjoint clusters by minimizing intra-class inertia, followed by a post-processing filtering step that eliminates clusters below a minimum size threshold to remove high-frequency noise from the final mask.

III. Experiments

The Q4S framework is evaluated using a subset of the CocoStuff dataset [13], selected to encompass a wide variety of semantic classes, scene types, and lighting conditions. The workflow consists of two main phases: a self-supervised pretraining stage using a classical backbone and a quantum semantic segmentation stage simulated via the PennyLane library [14], with detailed configurations provided in Table I. To satisfy the quantum amplitude embedding requirements, the number of qubits is directly determined by the output dimensionality of the final backbone layer. Ground truth masks are reserved strictly for performance evaluation; during this phase, the Hungarian algorithm [15] is employed to optimally map predicted clusters to ground truth segments before computing the final metrics. The classical baseline is a single linear layer ($512 \to 49$, tanh-bounded), matching the 7×7 spatial dimension of ResNet-18's final layer rather than the raw qubit count, with depth restricted to one layer to mirror $L = 1$.

TABLE I
HYPERPARAMETERS AND ARCHITECTURAL DETAILS

Phase	Parameter	Value
Pretraining	Backbone	ResNet-18
	Projection Head	Linear (512-256-128)
	Loss Function	SimCLR
	Optimizer	Adam ($lr = 1e^{-3}$)
	Batch Size / Epochs	64 / 200
	Temperature (τ)	0.1
Segmentation	Quantum Expander	Linear (9-512)
	Wires (n)	9
	Regularization (b)	0.1
	Layers (L)	1
	Epochs	300
	Clustering (K)	12

IV. RESULTS & DISCUSSION

The Q4S framework demonstrates robust semantic representation capabilities across the dataset (Table II). This performance is primarily driven by the model's over-clustering mechanism, which effectively captures intra-class variance by allowing more clusters than ground-truth classes. While global object shapes are accurately localized, as supported by high mIoU and F1 scores, the model exhibits a noticeable trade-off between high-level semantic extraction and low-level spatial precision.

TABLE II
Q4S SEMANTIC SEGMENTATION RESULTS ON A COCOSTUFF SUBSET OF 100 IMAGES (MEAN $\pm$ STD). WE EXPLAIN THIS DATASET SIZE DUE TO HIGH SIMULATION COST FOR OUR SINGLE IMAGE PROCESS PIPELINE.

Metric	Classical Baseline	Q4S
mIoU	$\mathbf{0.5933 \pm 0.1754}$	0.5827 ± 0.1806
F1	$\mathbf{0.6913 \pm 0.1761}$	0.6774 ± 0.1838
Pixel Acc.	0.8506 ± 0.1042	$\mathbf{0.8629 \pm 0.0741}$
MPA	0.7862 ± 0.1543	$\mathbf{0.7972 \pm 0.1496}$

Throughout the training epochs, the framework successfully learns to group semantically similar pixels. However, it encounters limitations in boundary adherence, leading to blurry contours and edge misclassifications (Fig 3). This constraint stems from the fact that pixels within the same spatial region share near-identical features. This spatial artifacting is particularly evident when using nearest-neighbor resizing, which exposes the 7×7 grid of the ResNet-18 backbone's final layer. While bilinear resizing smooths these boundaries, it leaves them blurry, and bicubic resizing improves sharpness at the expense of small-structure precision.

To change this, future work will focus on integrating boundary refinement modules and leveraging earlier layers of the pretrained backbone. In architectures like ResNet-18, earlier layers preserve higher spatial diversity with smaller feature sizes, allowing for enhanced edge precision without increasing the qubit requirement. This strategy relies on self-supervised information parallelization, which offers broad applicability across various domains. By parallelizing discrete spatial regions and correlating the quantum projection with the backbone's features, Q4S models complex relations with a limited number of qubits. This algorithmic design avoids the need to process the entire image simultaneously, ensuring scalability to larger images via region subdivision.

Finally, comparing our results (Table II) to a classical baseline of equivalent dimensionality reveals complementary strengths: the baseline scores higher on mIoU and F1, while Q4S scores higher on Pixel Accuracy and MPA. Given the reported standard deviations, none of these differences are statistically significant, confirming Q4S performs on par with its classical counterpart. Its edge on MPA, unlike Pixel Accuracy, is not dominated by majority classes and may suggest better representation of minority classes, though per-class analysis would confirm it. This result supports our goal, not quantum superiority but a self-supervised segmentation methodology that, to our knowledge, is the first to combine self-supervised pixel-level learning with a variational segmentation head, positioning Q4S as a feasibility baseline rather than a performance benchmark.

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